

label-and-one-hot-encoding

1 Label and Onehot Encoding

```
[1]: import pandas as pd
      from sklearn.preprocessing import LabelEncoder, OneHotEncoder
```

```
[2]: # Step 1: Load the dataset
      df = pd.read_csv("encoding.csv")
```

```
[3]: # Step 2: Apply Label Encoding (Converting categorical columns into numeric
      ↪ labels)
      label_enc = LabelEncoder()
      df['Color_LabelEncoded'] = label_enc.fit_transform(df['Color'])
      df['Size_LabelEncoded'] = label_enc.fit_transform(df['Size'])
```

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[4]: df
```

```
[4]:
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	ID	Color	Size	Price	Color_LabelEncoded	Size_LabelEncoded
0	1	Red	Small	10.5	2	2
1	2	Blue	Medium	15.0	0	1
2	3	Green	Large	12.3	1	0
3	4	Blue	Small	14.2	0	2
4	5	Red	Large	13.1	2	0
5	6	Green	Medium	16.8	1	1
6	7	Red	Small	11.5	2	2
7	8	Blue	Large	17.3	0	0
8	9	Green	Medium	14.9	1	1
9	10	Red	Large	15.7	2	0

```
[8]: # Step 3: Apply One-Hot Encoding (Creating binary columns)
      df_onehot = pd.get_dummies(df, columns=['Color', 'Size'], prefix=['Color', 'Size'])
```

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[10]: df_onehot
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[10]:
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	ID	Price	Color_LabelEncoded	Size_LabelEncoded	Color_Blue	Color_Green	Color_Red	Size_Large	Size_Medium	Size_Small
0	1	10.5	2	2	False	False	True	False	False	True
1	2	15.0	0	1	True	False	False	True	False	False

2	3	12.3	1	0	False	True
3	4	14.2	0	2	True	False
4	5	13.1	2	0	False	False
5	6	16.8	1	1	False	True
6	7	11.5	2	2	False	False
7	8	17.3	0	0	True	False
8	9	14.9	1	1	False	True
9	10	15.7	2	0	False	False

	Color_Red	Size_Large	Size_Medium	Size_Small
0	True	False	False	True
1	False	False	True	False
2	False	True	False	False
3	False	False	False	True
4	True	True	False	False
5	False	False	True	False
6	True	False	False	True
7	False	True	False	False
8	False	False	True	False
9	True	True	False	False

```
[11]: # Convert boolean values to integers
df_onehot = df_onehot.astype(int)
```

```
[12]: df_onehot
```

```
[12]:
```

	ID	Price	Color_LabelEncoded	Size_LabelEncoded	Color_Blue	Color_Green	\
0	1	10	2	2	0	0	
1	2	15	0	1	1	0	
2	3	12	1	0	0	1	
3	4	14	0	2	1	0	
4	5	13	2	0	0	0	
5	6	16	1	1	0	1	
6	7	11	2	2	0	0	
7	8	17	0	0	1	0	
8	9	14	1	1	0	1	
9	10	15	2	0	0	0	

	Color_Red	Size_Large	Size_Medium	Size_Small
0	1	0	0	1
1	0	0	1	0
2	0	1	0	0
3	0	0	0	1
4	1	1	0	0
5	0	0	1	0
6	1	0	0	1
7	0	1	0	0

8	0	0	1	0
9	1	1	0	0

```
[16]: print(df_onehot)
```

	ID	Price	Color_LabelEncoded	Size_LabelEncoded	Color_Blue	Color_Green	\
0	1	10	2	2	0	0	
1	2	15	0	1	1	0	
2	3	12	1	0	0	1	
3	4	14	0	2	1	0	
4	5	13	2	0	0	0	
5	6	16	1	1	0	1	
6	7	11	2	2	0	0	
7	8	17	0	0	1	0	
8	9	14	1	1	0	1	
9	10	15	2	0	0	0	

	Color_Red	Size_Large	Size_Medium	Size_Small
0	1	0	0	1
1	0	0	1	0
2	0	1	0	0
3	0	0	0	1
4	1	1	0	0
5	0	0	1	0
6	1	0	0	1
7	0	1	0	0
8	0	0	1	0
9	1	1	0	0

```
[14]: from IPython.display import display
display(df_onehot)
```

	ID	Price	Color_LabelEncoded	Size_LabelEncoded	Color_Blue	Color_Green	\
0	1	10	2	2	0	0	
1	2	15	0	1	1	0	
2	3	12	1	0	0	1	
3	4	14	0	2	1	0	
4	5	13	2	0	0	0	
5	6	16	1	1	0	1	
6	7	11	2	2	0	0	
7	8	17	0	0	1	0	
8	9	14	1	1	0	1	
9	10	15	2	0	0	0	

	Color_Red	Size_Large	Size_Medium	Size_Small
0	1	0	0	1
1	0	0	1	0
2	0	1	0	0

3	0	0	0	1
4	1	1	0	0
5	0	0	1	0
6	1	0	0	1
7	0	1	0	0
8	0	0	1	0
9	1	1	0	0